Wstawki asemblerowe

Debug x86 Lokalny debuger Windows

ConsoleApplication1.cpp

ConsoleApplication1 (Globalny zasięg)

```cpp
#include <iostream>
using namespace std;
int wynik = 0;
int k = 4;
int x = 2;
int main()
{
    _asm {
        xor edx,edx
        mov eax,1
        mov ecx,k
        mov ebx,x
    et1:
        mul bx
        loop et1
        mov wynik,eax
    }
    cout << "Wartosc " << x << "^"<< k <<" =" << wynik << endl;
}
```

Konsola debugowania progra

```
Wartosc 2^4 =16
```

Wywołanie funkcji z pliku *.asm x86

Debug | x86 | Lokalny debuger Windows

ConsoleApplication1.cpp | x_do_y.asm

ConsoleApplication1 | (Globalny zasięg)

```cpp
#include <iostream>
using namespace std;
extern "C" int __stdcall  x_do_y(int x, int y);

int main()
{
    cout << "Zad 1" << endl;
    int x = 2;
    int y = 4;
    int wynik = x_do_y(x,y);
    cout << "Wartosc " << x << "^" << y << " =" << wynik << endl;
}
```

Konsola debugowania progra

```
Zad 1
Wartosc 2^4 =16
```

x_do_y.asm | ConsoleApplication1.cpp

```asm
.486
.model flat, stdcall
.stack 4096
PUBLIC x_do_y
.data
.code
ALIGN 8
x_do_y proc, Arg1:DWORD,Arg2:DWORD
;in      x Arg1  y Arg2
;out     wynik w EAX
         mov ebx,Arg1
         mov ecx,Arg2
         xor edx,edx
         mov eax,1
      et1:
         mul bx
         loop et1
     ret;
x_do_y endp
end
```

Wywołanie funkcji z pliku *.asm x64

Debug | x64 | Lokalny debuger Windows

ConsoleApplication1.cpp

ConsoleApplication1 | (Globalny zasięg)

```cpp
#include <iostream>
using namespace std;
extern "C" int __stdcall  x_do_y(int x, int y);

int main()
{
    cout << "Zad 1-x64" << endl;
    int x = 2;
    int y = 4;
    int wynik = x_do_y(x,y);
    cout << "Wartosc " << x << "^" << y << " =" << wynik << endl;
}
```

x_do_y.asm | ConsoleApplication1.cpp

```asm
.code
ALIGN 8
x_do_y proc
;in      x Arg1 RCX y Arg2 RDX
;out     wynik w EAX
         xor rax,rax
         xor rbx,rbx
         mov rbx,rcx
         mov rcx,rdx

         mov rax,1
      et1:
         mul ebx
         loop et1
      ret;
x_do_y endp
end
```

Konsola debugowania progra

```
Zad 1-x64
Wartosc 2^4 =16
```

Czysty asembler

x86, x64

X64 – wyświetlanie wartości całkowitych

Debug x64 Lokalny

ie zakończona. Może pomóc wyłączenie rozszerzenia A

main.asm

```
1  includelib ucrt.lib
2  includelib legacy_stdio_definitions.lib
3  includelib kernel32.lib
4  includelib user32.lib
5
6  option  casemap:none
7
8  EXTERN printf: PROC
9
10 .data
11      fmtstr_w db "wynik=%d",10,0
12      i8  db 13      ;8 bitowa
13      i16 word 14    ; 16 bitowa
14      i32 dd 15      ;32 bitowa
15      i64 dq 16      ;64 bitowa
16
17 .code
18 main proc
19
20     sub rsp, 56
21     ; liczna 64 bitowa
22     lea rcx,fmtstr_w
23     mov rdx, i64
24     call printf
25
26     ; liczna 32 bitowa
27     lea rcx,fmtstr_w
28     xor rdx,rdx
29     mov edx, i32
30     call printf
31     ; liczna 16 bitowa
32     lea rcx,fmtstr_w
33     xor rdx,rdx
34     mov dx, i16
35     call printf
36     ; liczna 8 bitowa
37     lea rcx,fmtstr_w
38     xor rdx,rdx
39     mov dl, i8
40     call printf
41
42     mov rax,0
43     add rsp, 56
44     ret
45
46 main endp
47 end
```

Konsola debugowania progra

```
wynik=16
wynik=15
wynik=14
wynik=13
```

Debug x64 Lokalny

ie zakończona. Może pomóc wyłączenie rozszerzenia A

main.asm

```
1  includelib ucrt.lib
2  includelib legacy_stdio_definitions.lib
3  includelib kernel32.lib
4  includelib user32.lib
5
6  option  casemap:none
7
8  EXTERN printf: PROC
9
10 .data
11      fmtstr_w db "wynik=%d",10,0
12      i8  db 13      ;8 bitowa
13      i16 word 14    ; 16 bitowa
14      i32 dd 15      ;32 bitowa
15      i64 dq 16      ;64 bitowa
16
17 .code
18 main proc
19
20     sub rsp, 56
21     ; liczna 64 bitowa
```

X64 – wyświetlanie wartości całkowitych

Debug x64 Loka

ie zakończona. Może pomóc wyłączenie rozszerzen

main.asm

```
1   includelib ucrt.lib
2   includelib legacy_stdio_definitions.lib
3   includelib kernel32.lib
4   includelib user32.lib
5
6   option  casemap:none
7
8   EXTERN printf: PROC
9
10  .data
11      fmtstr_w db "wynik=%d",10,0
12      i8  db 13       ;8 bitowa
13      i16 word 14     ; 16 bitowa
14      i32 dd 15       ;32 bitowa
15      i64 dq 16       ;64 bitowa
16
```

```
17  .code
18  main proc
19
20      sub rsp, 56
21      ; liczna 64 bitowa
22      lea rcx,fmtstr_w
23      mov rdx, i64
24      call printf
25
26      ; liczna 32 bitowa
27      lea rcx,fmtstr_w
28      xor rdx,rdx
29      mov edx, i32
30      call printf
31      ; liczna 16 bitowa
32      lea rcx,fmtstr_w
33      xor rdx,rdx
34      mov dx, i16
35      call printf
36      ; liczna 8 bitowa
37      lea rcx,fmtstr_w
38      xor rdx,rdx
39      mov dl, i8
40      call printf
41
42      mov rax,0
43      add rsp, 56
44      ret
```

Konsola debugowania progra

```
wynik=16
wynik=15
wynik=14
wynik=13
```

x86 – wyświetlanie wartości całkowitych

Debug | x86 | Lokalny debuger Windows | Auto

main.asm

```
.686
.model flat, stdcall
.stack 4096

include \masm32\include\msvcrt.inc
includelib \masm32\lib\msvcrt.lib

printf PROTO C, :VARARG


ExitProcess PROTO, dwExitCode:DWORD

.data
fmt db "%s %u", 10, 0                   ; '10': printf of msvcrt.dll doesn't accept "\n"
nagl db "Wynik: ", 0

.code
main PROC
    mov eax,12
    call output                         ; "Message from ASM ..."


    INVOKE ExitProcess, eax
main ENDP

output PROC                             ; ARG: EAX
    invoke printf, OFFSET fmt, OFFSET nagl, eax
    ret
output ENDP

END
```

Konsola debugowania progra

```
Wynik:  12

C:\vs2019_c\arch_komp\x86\
```

x86 – wyświetlanie wartości całkowitych

Debug | x86 | Lokalny debuger Windows | Auto

main.asm

```
.686
.model flat, stdcall
.stack 4096

include \masm32\include\msvcrt.inc
includelib \masm32\lib\msvcrt.lib

printf PROTO C, :VARARG


ExitProcess PROTO, dwExitCode:DWORD

.data
fmt db "%s %u", 10, 0                   ; '10': printf of msvcrt.dll doesn't accept "\n"
nagl db "Wynik: ", 0

.code
main PROC
    mov eax,12
    call output                         ; "Message from ASM ..."


    INVOKE ExitProcess, eax
main ENDP

output PROC                             ; ARG: EAX
    invoke printf, OFFSET fmt, OFFSET nagl, eax
    ret
output ENDP

END
```

Konsola debugowania progra

```
Wynik:  12

C:\vs2019_c\arch_komp\x86\
```

x86 – wyświetlanie wartości całkowitych

Debug | x86 | Lokalny debuger Windows | Auto

main.asm

```
.686
.model flat, stdcall
.stack 4096

include \masm32\include\msvcrt.inc
includelib \masm32\lib\msvcrt.lib

printf PROTO C, :VARARG


ExitProcess PROTO, dwExitCode:DWORD

.data
fmt db "%s %u", 10, 0                   ; '10': printf of msvcrt.dll doesn't accept "\n"
nagl db "Wynik: ", 0

.code
main PROC
    mov eax,12
    call output                         ; "Message from ASM ..."


    mov eax,3
    mov ecx,4
    push 2
    call dodaj
    add esp,4

    call output                         ; "Message from ASM ..."

    INVOKE ExitProcess, eax
main ENDP

output PROC                             ; ARG: EAX
    invoke printf, OFFSET fmt, OFFSET nagl, eax
    ret
output ENDP

dodaj proc              ;2*arg+EAX
    mov ebx,[esp+4]
    imul ebx,ecx
    add eax,ebx
    ret
dodaj endp
END
```

Konsola debugowania progra

```
Wynik:   12
Wynik:   11
```

x86 – wyświetlanie wartości rzeczywistych

Irvine32.inc | main.asm

```
; Include file for Irvine32.lib            (Irvine32.inc)

;OPTION CASEMAP:NONE        ; optional: make identifiers case-sensitive

INCLUDE SmallWin.inc        ; MS-Windows prototypes, structures, and constants
INCLUDE VirtualKeys.inc

.NOLIST
; Last update: 7/29/05

;-------------------------------------
; Procedure Prototypes
;-------------------------------------
CloseFile PROTO     ; close a file handle
Clrscr PROTO        ; clear the screen
CreateOutputFile PROTO      ; create file for writing
Crlf PROTO      ; output carriage-return / linefeed
Delay PROTO     ; delay for n milliseconds
DumpMem PROTO       ; display memory dump
DumpRegs PROTO      ; display register dump
GetCommandTail PROTO        ; get command-line string
GetDateTime PROTO,      ; get system date and time
  startTime:PTR QWORD
GetMaxXY PROTO      ; get console rows and columns
GetMseconds PROTO       ; get milliseconds past midnight
GetTextColor PROTO      ; Get the console window's color attributes.
Gotoxy PROTO        ; set the cursor position
IsDigit PROTO       ; return ZF=1 if AL is a decimal digit
MsgBox PROTO        ; display popup message box
MsgBoxAsk PROTO     ; display popup yes/no question box
OpenInputFile PROTO     ; open file in input mode
ParseDecimal32 PROTO        ; convert unsigned decimal string to 32 bit binary
```

stackoverflow | Products | OverflowAI | Search...

Home NEW
Questions
Tags
Saves
Discussions Labs
Chat
Articles
Users
Companies
COLLECTIVES +
Communities for your favorite technologies.
Explore all Collectives

2 Answers

Sorted by: Highest score (default)

1

I will suppose you're using masm32 as your assembler, to properly set up your MASM project with Irvine32 in Visual Studio, follow these steps:

Download and Prepare Irvine32 Library:

- Download Irvine32 library from https://github.com/meixinchoy/Irvine-library.
- Unzip and move the folder to C:\Irvine.

Check Build Customization:

- Right-click your project in Visual Studio, select Build Dependencies > Build Customizations.
- Ensure the MASM option (.targets, .props) is checked.

Build Customizations

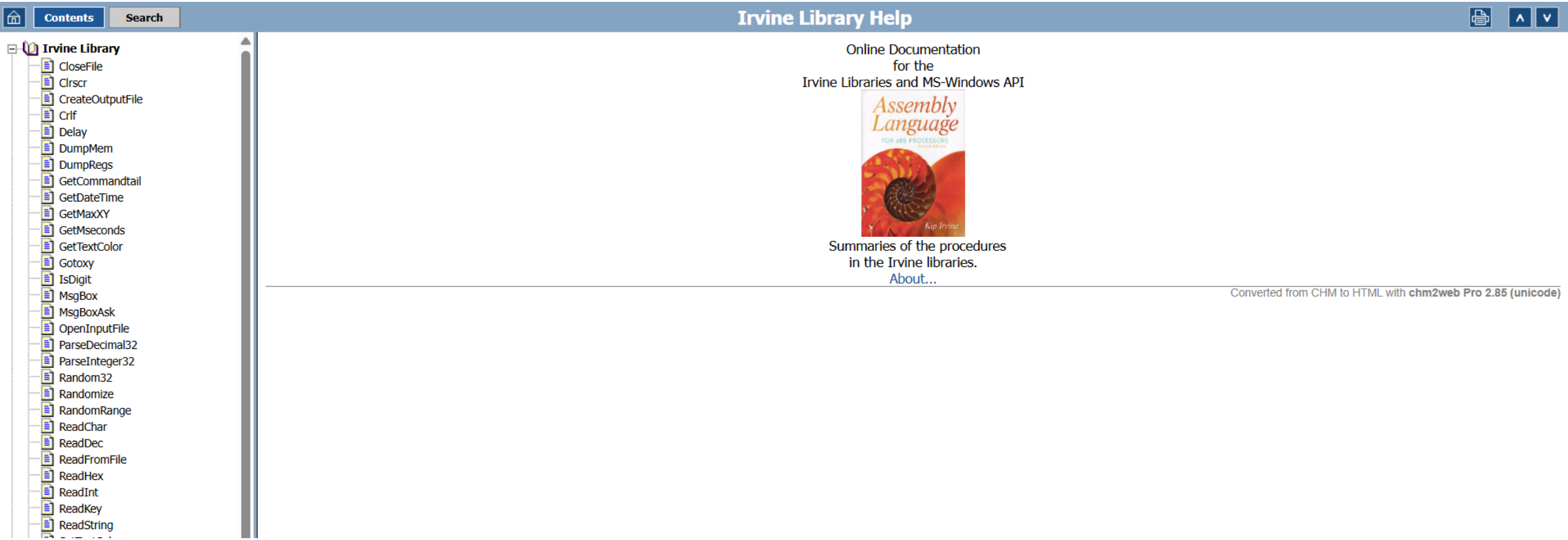
Set .asm File Properties:

- In Solution Explorer, right-click your .asm file, choose Properties and confirm Item Type is set to Microsoft Macro Assembler.

https://stackoverflow.com/questions/78174941/how-to-install-and-setup-irvine-library-in-vs-code-2019-for-assembly-language

x86 – wyświetlanie wartości rzeczywistych

https://csc.csudh.edu/mmccullough/asm/help/



x86 – wyświetlanie wartości rzeczywistych

Debug x86

Irvine32.inc main.asm

```
include Irvine32.inc

.data
    wart real4 1.7
    wart2 real4 0.11
.code
main proc
    finit
    fld wart
    fadd wart2
    call writefloat
    exit

main endp
end
```

Konsola debugowania progra

```
+1.8100000E+000
C:\vs2019_c\arch_komp\x86\czyst
```